West Bengal's graduates and formal jobs: what public data can and cannot say

A response to a public question on jobs created, joining dates and planned migration, with the wider picture the question did not ask for

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Built entirely from public data: all figures re-derived from source files and independently validated (264 checks)

Public data shows thin formal job entry that West Bengal shares with several states; it does not show how many graduates are unplaced or leave

- **Public data answers the question only in part.** It measures formal job entry. It does not give company-level joining dates, a count of graduates who leave, or a state target. Treat every figure here as a directional indicator, not a decision-grade estimate.
- **Jobs created:** West Bengal recorded 458,000 net new formal-sector workers in 2024-25, 212,000 of them aged 18 to 25. Formal work only; it includes people who left and rejoined, so it is not a count of first jobs.
- **The graduate gap cannot be read as unplaced graduates.** Raw, the state's institutions produced 1.89 million undergraduate graduates over five years against 0.89 million formal entrants aged 18 to 25. The two counts differ in place, age, kind and period. Once further study and non-participation are allowed for, the remainder is between −0.05 and +0.54 million depending on the age band used, and it could be informal or government work, non-graduate entrants, or moves out of state.
- **The stronger finding needs no graduate assumption:** per 100 young people, Tamil Nadu, Telangana, Karnataka and Maharashtra record 9.7 to 11.1 formal entrants a year, West Bengal 2.1 and its four neighbours 0.3 to 1.9. A household survey orders the states the same way, which supports the pattern but not EPFO's exact levels.
- **Job quality:** 60.5% of West Bengal's regular non-farm jobs carry no social security (India 51.7%). Graduate unemployment is 11.4%, close to India's 11.2%, and about 40% of women graduates are in the labour force.
- **No state target for stopping migration was found,** but the state's budget speeches, policies and summit papers have not been read, so the question stays open. Pages 16 and 17 set out a graduate tracking study and three defined indicators that would give decision-grade answers.

Source: AISHE final reports 2019-20 to 2023-24; EPFO provisional payroll estimates (20 Jan and 20 Sep 2025); PLFS 2025 annual report, district snapshot and million-plus cities report; team analysis

Public data answers the question in part: it measures formal job entry and a graduate gap, not joining dates, migration or targets

The question asked how many jobs have been created in West Bengal, when people join, and how many migrations are planned to be stopped over two years, given the graduates the state produces. Exhibit 1 sets each part against what public sources can offer.

Exhibit 1: The three questions against public data

Question asked	Closest public measure	What it can show	Status
How many jobs have been created?	EPFO net new formal payroll by state and age band	458,000 in 2024-25, 212,000 aged 18 to 25. Formal work only; includes people who rejoined, so not a count of first jobs	**Partly answered**
When do people join?	EPFO monthly totals by state	Monthly totals for Apr to Nov 2024 and Apr to Jul 2025. No company-level dates found	**Partly answered**
How many migrations are planned to be stopped?	None: no public series counts graduates who leave; no state target found	A gap of 0.28 to 1.0 million between graduates and formal entry, which is not a migration count	**Not answerable; target search incomplete**

Part 1 · Your three questions

Pages 4 to 6 take jobs created, joining dates and migration in turn, with what the data can and cannot say.

Part 2 · What you did not ask

Pages 7 to 17 cover the size of the gap and how far it can be read, how West Bengal compares, job quality, graduates, Kolkata, and a design that would settle the question.

Appendices

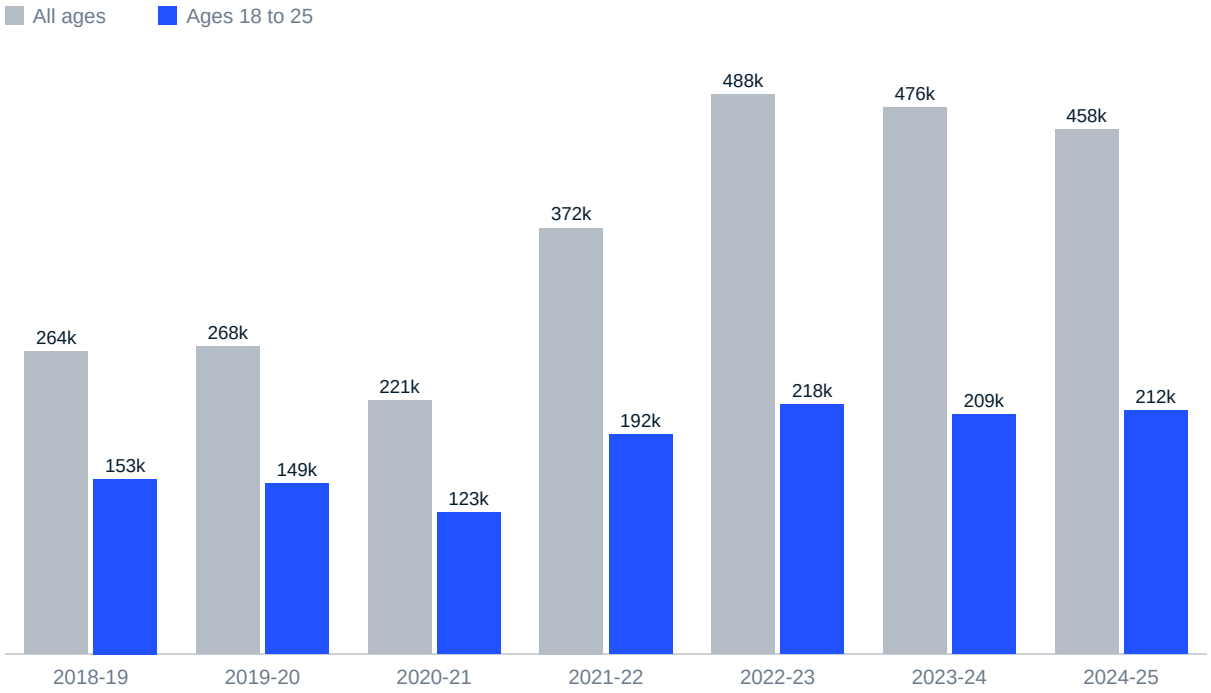
Sources, limits, and how every figure was validated before this report was produced.

Source: EPFO provisional payroll estimates; AISHE final reports; team analysis

West Bengal's net formal payroll additions peaked at 488,000 in 2022-23 and have eased to 458,000 in 2024-25

Exhibit 2: Net new formal-sector payroll, West Bengal

People per financial year (April to March)



What the figure is, and is not

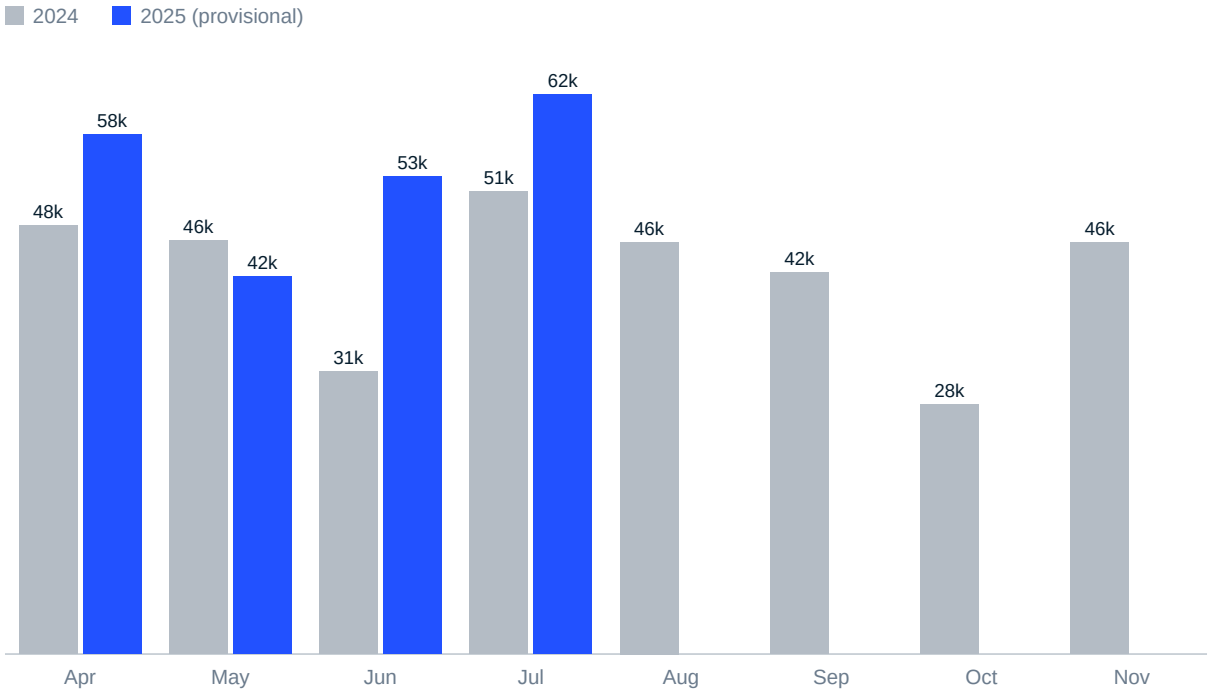
- The 18 to 25 age group has stayed between 209,000 and 218,000 for three years.
- It counts new provident-fund members, less those who left, plus those who rejoined.
- It covers the formal sector only. Rejoiners are included, so it is not a count of first jobs and can overstate new ones. Informal work is not counted.
- For context, regular wage or salaried work makes up 22.3% of West Bengal's workers (India 23.6%).

Source: EPFO provisional monthly payroll estimates, releases of 20 Jan and 20 Sep 2025 (state tables); PLFS 2025 annual report, Table 19; team analysis

On provisional figures, April to July 2025 added about 22% more net formal payroll than a year earlier; no company-level joining dates were found

Exhibit 3: Net new formal-sector payroll by month, West Bengal

People per month, all ages



Reading the months

- April to July 2025 added 214,690 against 176,091 in the same months of 2024.
- The comparison is not like for like: 2025 months have had less time to be updated, and both are provisional.
- October 2024 is the low month in West Bengal, as it is nationally.
- The two releases used do not cover December 2024 to March 2025 month by month.
- Monthly totals by state and age band are the finest breakdown in the releases reviewed. No company-level joining dates were found.

Source: EPFO provisional monthly payroll estimates, releases of 20 Jan 2025 (Apr to Nov 2024) and 20 Sep 2025 (Apr to Jul 2025); team analysis

Public data does not count graduates who leave, and no state target was found, though the search is incomplete

What public data offers

- A gap between graduates produced and formal job entry: 0.28 to 1.0 million over five years, and why it cannot be read as unplaced graduates (pages 7 and 8)
- Survey measures of unemployment and of young people not working or studying
- Occasional national migration surveys exist; they were not reviewed for this report and are not specific to graduates

What is missing

- A public count of graduates who leave the state
- A state target. The budget speeches, IT and industrial policies and summit papers could not be read for this report, so we can neither confirm nor deny that a white paper exists

What a target would need

- A definition: leaving for study or for work, for how long, and which graduates
- A baseline measured the same way each year
- Linked records from graduates' institutions to later employment

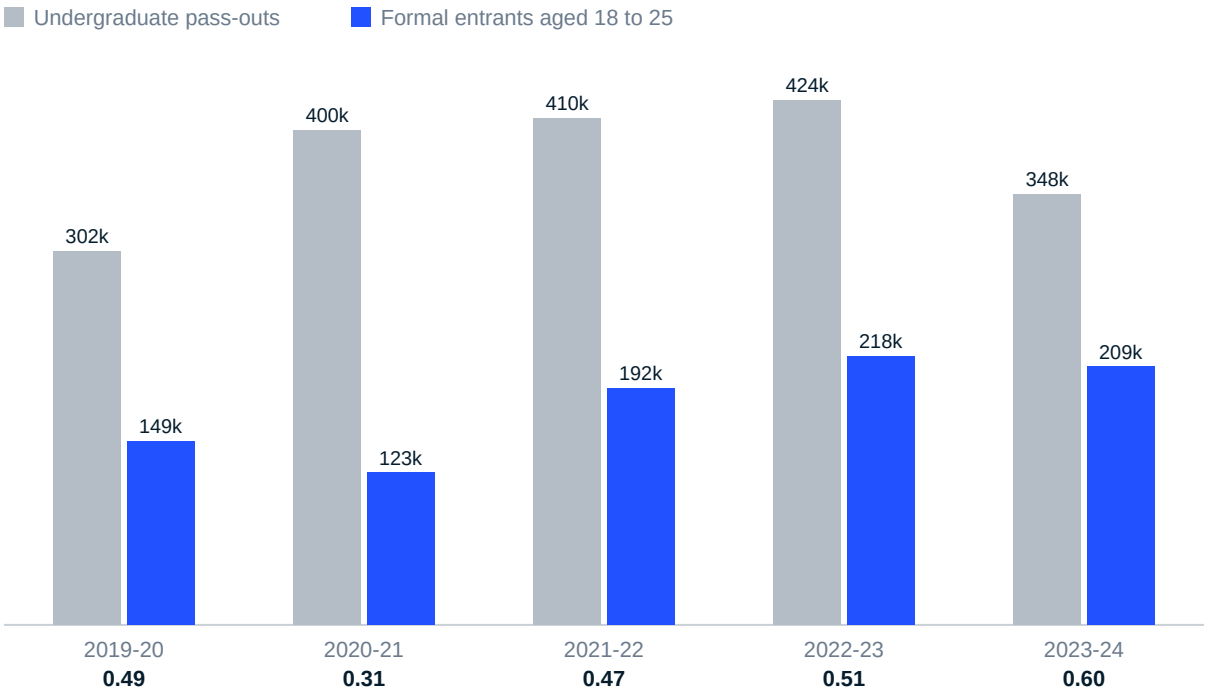
Why this matters. Graduates who study further, work informally or for government, or are out of work all sit inside the gap. Without a definition, any number put on "migrations stopped" cannot be checked.

Source: team analysis; AISHE, EPFO and PLFS as cited on the following pages

Counted raw, formal entry by 18 to 25-year-olds is below graduate output every year, but the two counts are not like for like

Exhibit 4: Undergraduate pass-outs and formal entrants, West Bengal

People per year. The figure under each year is formal entrants per pass-out



Raw ratio, before any adjustment

The raw ratio under four assumptions	Entrants per graduate
Baseline: ages 18 to 25 against all pass-outs	**0.47**
Graduate-age band only (22 to 25)	**0.26**
One-year lag in joining	**0.51**
Pass-outs counted only if in the labour force	**0.76**

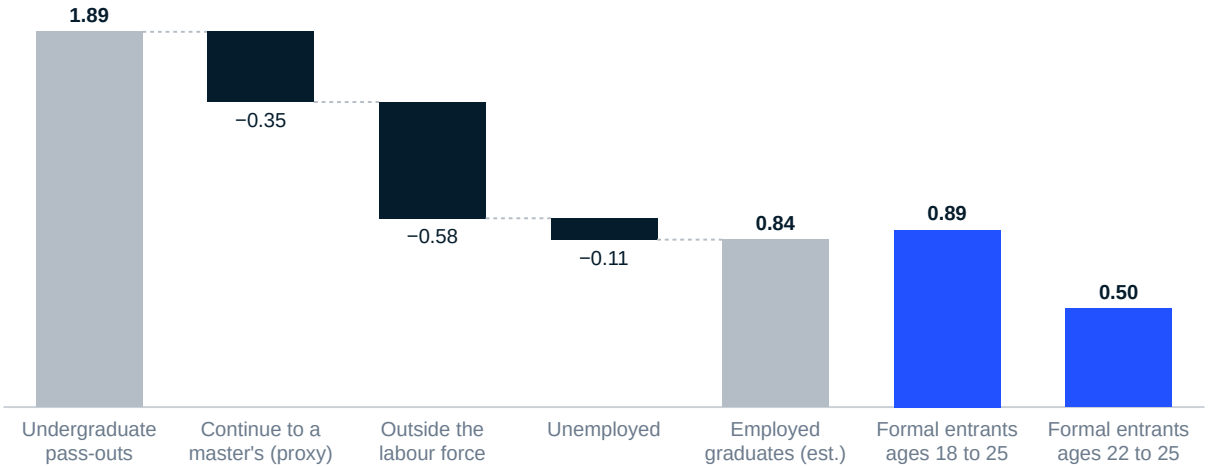
- Over five years: **1,885,522** pass-outs against **890,027** formal entrants.
- The raw difference is **0.28 to 1.0 million**. The next page shows why it cannot be read as unplaced graduates.
- Pass-outs for 2023-24 fall 18% for reasons the data does not explain, so five-year totals are used.

Note: AISHE academic years are matched to EPFO financial years by label; graduates mostly join the year after. Source: AISHE final reports 2019-20 to 2023-24 (Table 33); EPFO payroll estimates (state tables); PLFS 2025 (graduate participation by sex); team analysis

Allowing for further study and non-participation, the remainder is −0.05 to +0.54 million; it is not a count of unplaced graduates

Exhibit 5: From undergraduate pass-outs to employed graduates, West Bengal, five years

Millions of people, 2019-20 to 2023-24; illustrative, with the assumptions in the note below



Case	Employed graduates	Formal entrants	Remainder
Master's deducted, ages 18 to 25	0.84	0.89	**−0.05**
Master's deducted, ages 22 to 25	0.84	0.50	**+0.35**
No master's deduction, ages 18 to 25	1.03	0.89	**+0.14**
No master's deduction, ages 22 to 25	1.03	0.50	**+0.54**

Read with care: four mismatches

- **Place.** Pass-outs are counted by the institution's state; EPFO assigns workers to states by a method it does not state. Students, graduates and workers cross state lines.
- **Age.** Pass-outs are graduates of any age; payroll entrants are aged 18 to 25 and include people who are not graduates; population is aged 18 to 23.
- **Kind.** Payroll is a flow of new entrants. The household survey used to check it counts a stock of people in work.
- **Period.** Academic years run July to June, financial years April to March, and graduates mostly join the year after.

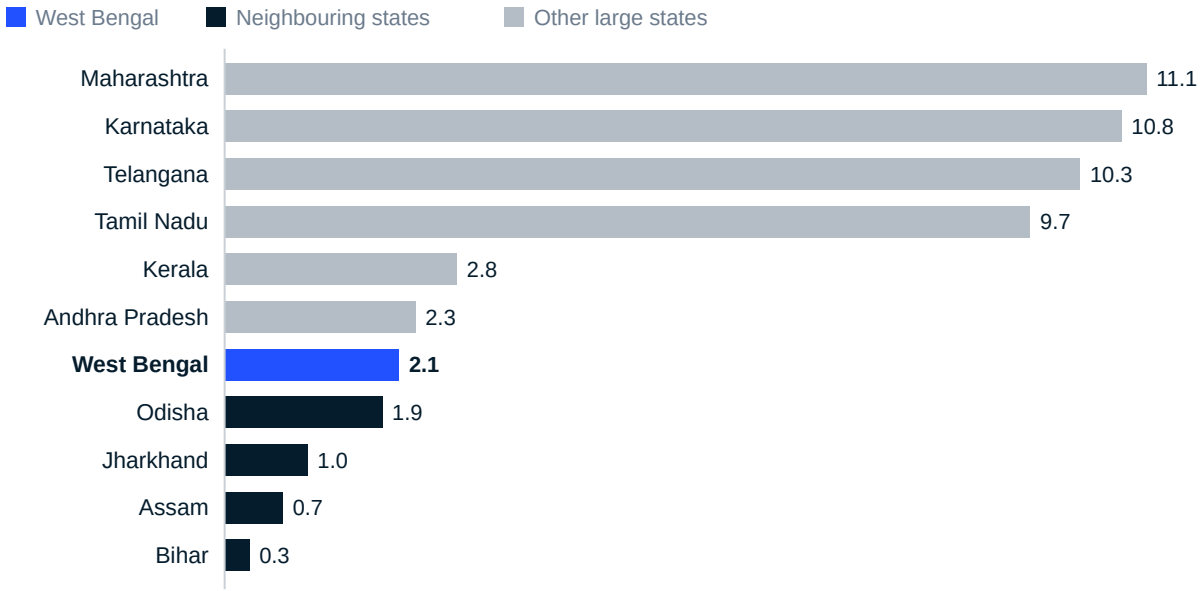
What the remainder could be. Informal or government work, formal entrants who are not graduates, people who moved in from other states, and graduates working in other states. This data cannot separate them.

Note: continuing to a master's is proxied by postgraduate pass-outs in the same five years (0.35 million, 19% of undergraduate pass-outs), which also include students from elsewhere. Participation by sex and the unemployment rate are the 2025 survey's graduate figures. Source: AISHE final reports (Table 33); EPFO payroll estimates; PLFS 2025 annual report, Tables 22 and 24; team analysis

Without any graduate assumption, four large states record 9.7 to 11.1 formal entrants per 100 young people and West Bengal 2.1; its neighbours record less

Exhibit 6: Net new formal payroll aged 18 to 25 per 100 people aged 18 to 23, 2023-24

Formal entrants a year per 100 young people



What stands out

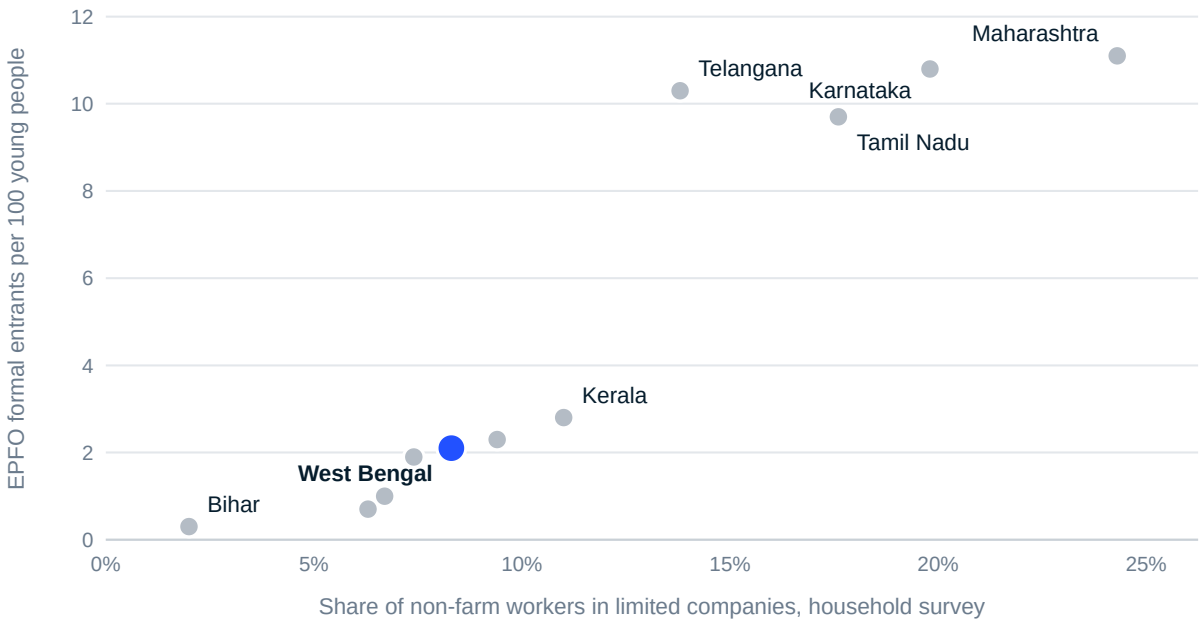
- West Bengal's four neighbours record 0.3 (Bihar) to 1.9 (Odisha).
- Andhra Pradesh (2.3) and Kerala (2.8) look like West Bengal on this measure; Kerala does not on the survey measure (next page).
- West Bengal's graduate output per young person (3.5 per 100) is 38% to 64% of the four large states' (5.5 to 9.1). Both parts of the ratio differ, but formal entry differs far more.
- Payroll entrants are aged 18 to 25 and the population 18 to 23. The same mismatch applies to every state, so the order is less affected than the levels.
- EPFO does not say how it assigns workers to states. If it uses the employer's registered state, Maharashtra's and Karnataka's figures could be inflated by head-office registration.

Source: EPFO provisional payroll estimates (Sep 2025 release, state tables); AISHE final report 2023-24 (Table 38, population aged 18 to 23); team analysis

A household survey orders the states the same way, which supports the pattern but cannot confirm EPFO's levels

Exhibit 7: EPFO formal entry against the survey's share of non-farm workers in limited companies

Eleven states with both measures



Does the pattern survive a different source?

	West Bengal	Four large states
Formal-job density per 100 people aged 15+	**5.1**	9.0 to 12.3
Non-farm workers in limited companies	**8.3%**	13.8% to 24.3%

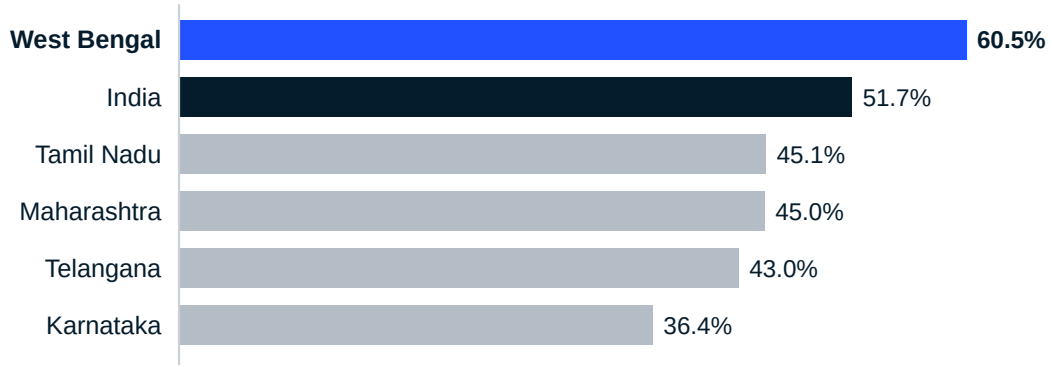
- EPFO counts a flow of new entrants; the survey counts a stock of people in work. Ordering the states alike does not show that EPFO's figures are accurate or comparable across states.
- Rank correlation with EPFO: **0.99** (limited-company share), **0.88** (formal-job density), **0.83** (regular salaried share). Eleven states, so indicative.
- West Bengal's density is **0.41 to 0.57** of the large states', a smaller gap than EPFO shows, though on a different kind of measure.
- Kerala is the exception: 40.0% of its workers are in regular salaried work, yet EPFO shows 2.8 entrants per 100. EPFO may understate its formal jobs.

Note: formal-job density = working-population ratio x regular wage or salaried share x share of regular jobs with some social security; rough, as the shares cover different groups of workers. Source: PLFS 2025 annual report, Tables 19, 23, 35, 36; EPFO payroll estimates; team analysis

West Bengal has a smaller share of regular salaried work than the large states, and more of it lacks social security

Exhibit 8: Regular non-farm jobs with no social security

Per cent of regular wage or salaried employees in non-farm work, 2025



Household survey, 2025	West Bengal	India	Four large states
Regular wage or salaried work, share of workers	**22.3%**	23.6%	27.1% to 37.3%
Non-farm workers in limited companies	**8.3%**	11.7%	13.8% to 24.3%

Reading the figures

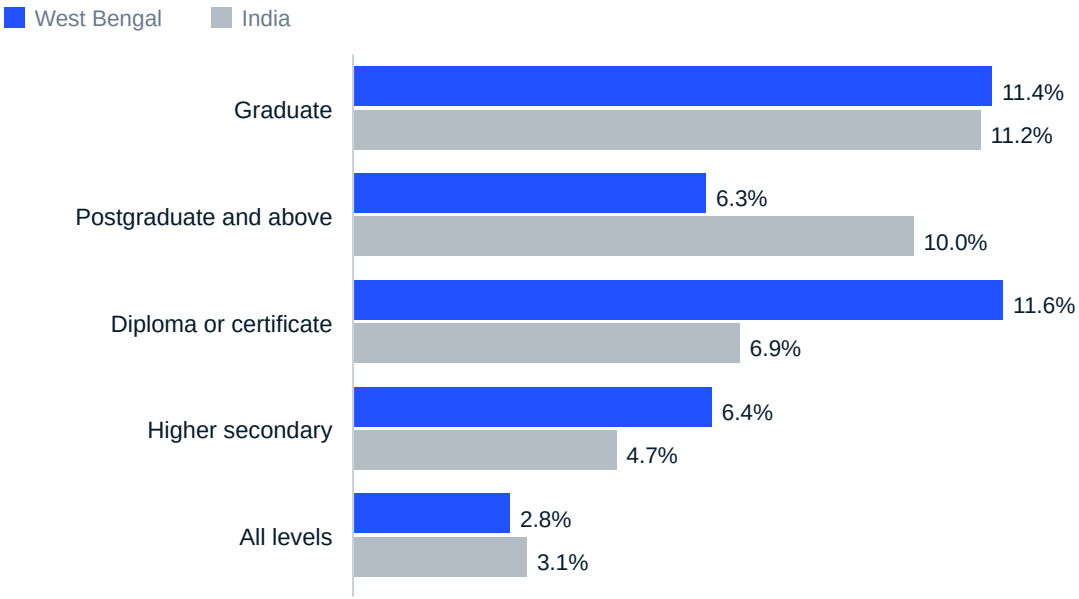
- 60.5% of West Bengal's regular non-farm jobs carry no social security, against 51.7% for India and 36.4% to 45.1% in the four large states. Only Bihar is higher among the states compared.
- No social security means not eligible for provident fund or pension, gratuity, or health care or maternity benefit (the survey's definition).
- Job count and job quality therefore need separate measures. Counting jobs alone would overstate progress.

Source: PLFS 2025 annual report, Tables 19, 35 and 36 (rural and urban, persons); team analysis

Graduate unemployment is four times the headline rate yet is close to India's; about 40% of women graduates are in the labour force

Exhibit 9: Unemployment rate by highest level of education

Per cent of the labour force, age 15 and over, 2025



What stands out

11.4%
graduates unemployed (all adults 2.8%)

40%
of women graduates in the labour force (men 86%)

- West Bengal is close to India for graduates (11.2%), so this is a national pattern. India's figures for women and men are 39% and 87%.
- Women are 52% of undergraduate pass-outs. Each ten points of extra participation among one year's women pass-outs would add about 18,000 people to the labour force.
- Diploma and certificate holders fare worse than India (11.6% against 6.9%). INFERENCE This may point to a mismatch in vocational training.
- These are sample estimates. The report publishes no relative standard error for these tables; a rough estimate puts them within about two points either side.

Note: 'Graduate' and 'postgraduate and above' are the survey's categories for highest level of education. Source: PLFS 2025 annual report, Tables 22 and 24 (usual status, rural and urban, persons); AISHE final report 2023-24 (Table 33); team analysis

Kolkata combines very low unemployment with the fifth-lowest regular pay among 46 million-plus cities

Exhibit 10: Average regular wage or salary, million-plus cities

Rupees a month, one dot per city (46 cities)

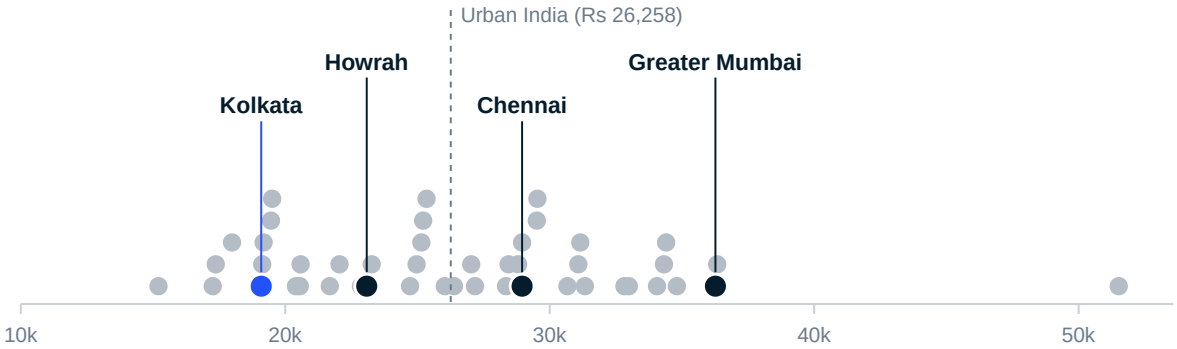
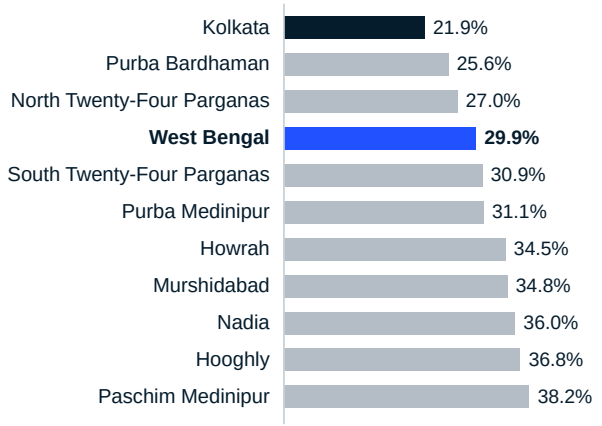


Exhibit 11: Youth not working or studying, by district

Per cent of 15 to 29-year-olds, 2025; ten most populous districts



- Kolkata averages Rs 19,093 a month; Howrah Rs 23,078 (rank 16 from the lowest); Greater Mumbai Rs 36,267; Chennai Rs 28,955.
- Kolkata's unemployment rate is 1.9% (relative standard error 28.8%, so approximate), against 4.9% for all million-plus cities. INFERENCE Low unemployment with low pay suggests many people take whatever work is available; cost of living and industry mix are other explanations.

Source: PLFS 2025 Labour Market Dynamics in Million-plus Cities (Tables 8 and 11, 13.1) and Labour Market Snapshot of Selected Districts; team analysis

Beyond the three questions, the evidence points to job creation and job quality more than to graduate supply

INFERENCE The points below are readings of the data, not findings, and each could be wrong. The last column says how firmly the data supports it.

Reading	Evidence	Confidence
The graduate-to-payroll gap is not a measure of unplaced graduates	After allowing for further study and non-participation the remainder is −0.05 to +0.54 million, depending on the age band used	High that it cannot be read that way
A migration target is mostly a job-creation target	Matching the lowest large state (9.7 per 100) would need about 4.6 times today's formal entry on the payroll measure, and 1.8 to 2.4 times on the survey measure	Medium
Job count and job quality need separate targets	60.5% of regular non-farm jobs carry no social security (India 51.7%)	High on the facts, medium on the policy reading
Women graduates are a large pool outside the labour force	40% participation against 86% for men; 52% of pass-outs are women; the pattern is national	High on the facts, medium on the policy reading
Vocational training deserves a look	Diploma and certificate unemployment 11.6% against India's 6.9%, a gap of 4.7 points, wider than the rough sampling error	Medium that the gap is real; low on the cause
Pay may matter more than job counts in Kolkata	Fifth-lowest regular pay of 46 cities with 1.9% unemployment	Low to medium: other explanations not ruled out
The people who leave may not be graduates	Not tested here; migration surveys would show lower-skilled out-migration better	Low
The shortfall is shared with other states	Andhra Pradesh and Kerala also record low formal entry per young person, and all four neighbours record fewer formal entrants per young person (0.3 to 1.9) than West Bengal (2.1)	Medium: payroll measure; Assam is level with West Bengal on the survey measure and Kerala is not low on it

Source: team analysis of AISHE, EPFO and PLFS data as cited on pages 4 to 13

Three actions would answer the original question directly

1. Track graduates after they pass out (design on page 16)

- EPFO accounts are linked to Aadhaar, and AISHE collects pass-out data from each institution
- A tracer study of where graduates work, for how much and in which state would replace most of the inferences in this report

2. Publish three numbers every year (definitions on page 17)

- Formal entry per 100 young people
- Graduate unemployment
- Median graduate pay

Any claim about jobs or migration could then be tested against them.

3. Close the open items

- Read the state's budget speeches, IT and industrial policies and summit papers for a stated target
- Obtain EPFO's method note on how workers are assigned to states
- Add factory employment from the Annual Survey of Industries

Exhibit 12: What each action resolves

Action	Resolves	What it needs
1. Track graduates after they pass out	Question 3: how many graduates leave, and where they work	Access to linked records held by the agencies concerned
2. Publish three numbers every year	Question 1, and the testing of any later claim	Formal entry and graduate unemployment use existing public data; graduate pay may need survey microdata
3. Close the open items	Question 3 (is there a target?) and how comparable the payroll figures are across states	The state's documents, EPFO's method note and the factory-survey file

Source: team analysis

A tracer study of West Bengal graduates would answer the question directly; this is a workable design

Exhibit 13: Proposed design for a graduate tracer study

Team proposal. Costs, legal basis and agreements between agencies have not been assessed.

Question it answers	Where do West Bengal's graduates work after they graduate: in which sector (formal, government, informal or self-employed), in which state and city, at what pay, and how many go on to further study or leave and return?
Who collects	The state's higher-education department and the universities hold institution records and contact details and run the survey. A statistical agency such as the NSO designs the sample. EPFO and ESIC verify formal employment under a data-sharing agreement. Roles are proposals, to be agreed.
Cohort	Undergraduate graduating classes of 2025 and 2026 from all institutions in the state. For scale, institutions reported 348,424 undergraduate pass-outs in 2023-24.
Sample	About 15,000 graduates per cohort, drawn at random and stratified by institution type, discipline, gender and district. This gives about ±1.0 percentage points at state level and about ±3.4 points for each of 12 subgroups of about 1,250 (95% confidence, design effect 1.5).
Follow-up	Contact at 6, 12 and 24 months after graduation; a new cohort every year.
Outcomes recorded	Activity (working, studying, unemployed, not seeking work); employer type and sector; contract and social-security status; monthly earnings band; district and state of work; whether the graduate lived or studied outside the state and why; plan to stay or move.
Linking and consent	Consent-based linkage to payroll records to confirm formal work. Data-protection rules and legal basis to be confirmed before fieldwork.
Publication	An annual state report with the three indicators on the next page, and anonymised data for researchers.

Note: margins of error use n = 1.96 x sqrt(design effect x p(1-p) / respondents) with p = 0.5 and a design effect of 1.5, an assumption. Source: AISHE final report 2023-24 (Table 33) for the cohort size; team design

Three yearly indicators would make claims testable, once each has a definition and a named data owner

Exhibit 14: Proposed indicators

Team proposal. Owners are the agencies that already produce the underlying data; whether each would publish the indicator is for them to decide.

Indicator	Definition	Data owner	Frequency	Known limits
Formal entry per 100 young people	Net new EPFO payroll, ages 18 to 25, in the financial year, divided by the population aged 18 to 23, times 100	EPFO (numerator); population projections as published in the higher-education survey tables, underlying source to be confirmed	Annual; the payroll is published monthly	Employer-state attribution unconfirmed; ages in numerator and denominator differ
Graduate unemployment rate	Persons whose highest education is graduate, aged 20 to 29 (and all graduates aged 15 and over, as published), unemployed as a share of the labour force, usual status	National Statistics Office, household survey (unit-level data for the age 20 to 29 cut)	Annual	Sample estimates; the published reports give no standard error for these tables
Median graduate pay	Median monthly earnings of regular wage or salaried employees with graduate education, aged 20 to 29, in rupees	National Statistics Office, household survey (unit-level data)	Annual	The reports reviewed give averages, not medians; small samples for a single state

Read together, the three give a yearly view of how many young people enter formal work, how graduates fare in the labour market, and what they earn. None of them counts migration, which is what the tracer study is for.

Source: team design; data owners as named in the sources on page 18

APPENDIX A

Sources and method

Topic	Source	Used for	Page
Graduates	All India Survey of Higher Education (Ministry of Education), final reports 2019-20 to 2023-24	Table 33 pass-outs by state and level; Table 38 population aged 18 to 23	aishe.gov.in/document-category/aishe-final-reports
Formal jobs	Employees' Provident Fund Organisation, provisional monthly payroll estimates, releases of 20 Jan and 20 Sep 2025	Net new payroll by state and age band, financial years 2018-19 to 2024-25; months Apr to Nov 2024 and Apr to Jul 2025	epfo.gov.in/data-hub
Employment	Periodic Labour Force Survey 2025 (MoSPI): annual report, district snapshot, million-plus cities report	Unemployment, participation, job type and social security by state; city and district figures	mospi.gov.in/publications-reports

- Graduate supply is undergraduate pass-outs counted by the state of the institution. Formal demand is EPFO net new payroll (new subscribers less exits plus rejoiners), ages 18 to 25 standing in for graduates.
- The gap is counted four ways (page 7) to show how far it moves with the assumptions.
- States are compared per 100 people aged 18 to 23, and checked against survey measures counted where people live (page 10).
- Files were downloaded on 4 and 5 November 2026. Code and cleaned data are in the project's state-jobs-gap repository.

Source: as listed

APPENDIX B

Limits and how each was handled

Limit	Effect	How it was handled
EPFO does not state how it assigns workers to states	Head-office states could be inflated	Order of states tested against survey data counted where people live; it holds
AISHE pass-outs for 2023-24 fall 18% in West Bengal	Single-year ratios are unreliable	Response rates do not explain it; five-year totals used
Academic and financial years differ	Joining lag blurs year-by-year matching	A one-year lag tested (ratio 0.51, against 0.47 baseline)
EPFO covers formal work only; net payroll is not first jobs	Overstates new jobs, ignores informal work	Described as a proxy throughout
Students and workers cross state lines	Distorts every ratio	Stated on each comparison; migration not inferred
Survey estimates carry sampling error	Small gaps may not be real	Rough error of about two points for graduate tables (our estimate; the report publishes none); small differences flagged
The raw graduate gap compares different kinds of counts	Cannot be read as unplaced graduates	Restated as a remainder after further study and non-participation (page 8); stronger indicator used
State documents not yet read	Question 3 cannot be settled	Stated as open

Source: team analysis

APPENDIX C

Validation: all 264 checks passed, and the pressure test corrected or tightened 16 points along the way

Check group	What was checked	Passed
AISHE	Pass-outs and population re-read from the raw workbooks	21 of 21
EPFO	West Bengal payroll re-read from the raw PDFs; two releases compared; definition checked	62 of 62
PLFS	Survey tables re-read by a second text route; table columns confirmed by viewing the tables	77 of 77
bridge		10 of 10
design		2 of 2
PLFS cities	City wages, rank and flags re-read from the raw report	9 of 9
PLFS districts	District figures re-read from the raw report	11 of 11
claim	Every derived figure, range, rank and correlation in the text recomputed	72 of 72

Figures in the text are inserted by code from the validated data. As a guard against typing errors, a final audit looked for each of the 363 numbers in the report body among the validated values and found 363; years, page and exhibit numbers are excluded. It checks that a number exists in the data, not that it is used correctly, which the claim checks above cover. The checks run before every build, and the report is not produced if any fails.

Source: report/validate.py and report/validation_log.csv in the project repository

Corrected during the pressure test

- A hand-copied value for Kolkata's unemployment carried a high-error flag that belongs to the women's rate; corrected in the data
- A draft headline said four states record 10 or more entrants per 100; Tamil Nadu records 9.7, so the range 9.7 to 11.1 is used
- 'Largest unused pool' for women graduates was not supported by the data; reworded to a large pool
- The payroll-based gap to the large states (2.1 against 9.7 to 11.1 per 100) is now shown beside the survey-based one (0.41 to 0.57 of theirs)
- An earlier inference that West Bengal is a net source of young workers was withdrawn; Kerala is not grouped with West Bengal on the survey measure
- The meaning of columns in three survey tables was confirmed by viewing the tables, not inferred from text layout
- The diploma-holder gap was first described as close to the sampling error; at 4.7 points it is wider than the rough error, so the wording and confidence were changed
- 'Regional' was changed to 'shared with other states', since Andhra Pradesh and Kerala also record low formal entry
- 'Joining dates are not public' was changed to 'none were found', since absence in the sources reviewed is not proof